
Oligonucleotide-directed mutagenesis using M13-derived vectors: an efficient and general procedure for the production of point mutations in any fragment of DNA

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ABSTRACT

This paper presents a versatile and efficient procedure for the construction of oligodeoxyribonucleotide directed site-specific mutations in DNA fragments cloned into M13 derived vectors. As an example, production of a transition mutation in a clone of the yeast *MATa1* gene is described. The oligonucleotide is hybridized to the template DNA and covalently closed double stranded molecules are generated by extension of the oligonucleotide primer with *E. coli* DNA polymerase (large fragment) and ligation with T4 DNA ligase. The resulting double stranded closed circular DNA (CC-DNA) is separated from unligated and incompletely extended molecules by alkaline sucrose gradient centrifugation. This purification is essential for production of mutants at high efficiency. Competent *E. coli* JM101 cells are transformed with the CC-DNA fraction and single stranded DNA is isolated from individual plaques. The recombinants are screened for mutant molecules by 1) restriction endonuclease screening for the loss of the *Hinf* I site in the target region, and 2) by dot blot hybridization using the mutagenic oligonucleotide as probe. Double stranded DNA is isolated from the mutant and the production of the desired mutation is verified by DNA sequencing. Efficiency of mutant production is in the range of 10-45% and no precautions to prevent mismatch repair are required.

INTRODUCTION

Oligonucleotide-directed mutagenesis provides a completely general method for producing defined point mutants (1). The technique resulted from the combination of a number of recent discoveries and observations about nucleic acids: 1) marker rescue of mutations in ϕ X174 by restriction fragments (2,3), 2) the stability of oligodeoxyribonucleotide-DNA duplexes containing mismatches (4-7), and 3) the ability of *E. coli* DNA polymerase in conjunction with T4 DNA ligase to extend oligonucleotide primers and to synthesize closed circular double stranded DNA molecules from single stranded templates (8,9). The basic principle (Figure 1) involves the enzymatic extension by *E. coli* DNA polymerase (large fragment) of an oligonucleotide primer hybridized to a single

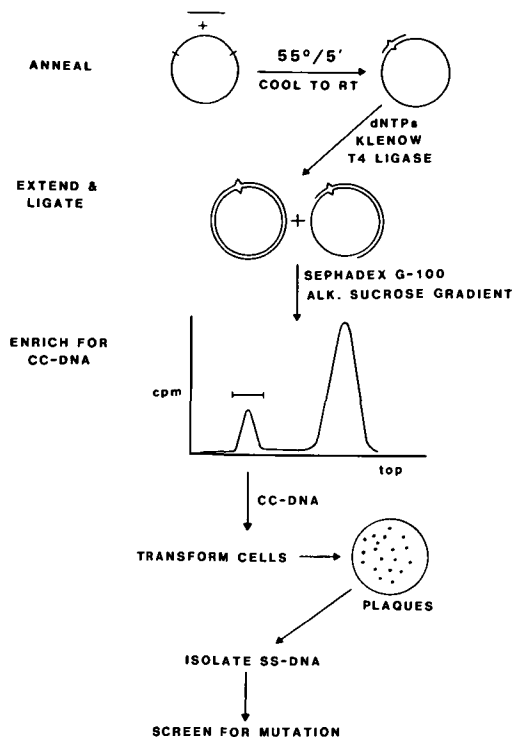


Figure 1. General scheme for oligonucleotide-directed mutagenesis.

stranded circular template. The oligonucleotide, 8-20 nucleotides, is complementary to a region of the template except for a mismatch that directs the mutagenesis. Closed circular double stranded molecules are formed by ligation of the newly synthesized strand with T4 DNA ligase and, in the present procedure, purified by alkaline sucrose gradient centrifugation. Upon transformation of competent cells with the *in vitro* synthesized closed circular DNA, a population of mutant and wild type molecules are obtained. Mutant molecules are distinguished from wild type by one of a number of screening procedures.

Earlier studies constructed a number of transition, transversion, and deletion mutations in ϕ 174 (10-12). Subsequently, the basic principle has been used to create specific mutations in DNA fragments cloned into pBR322 (13,14), fd (15), and M13 (16-21) vectors, but at much lower efficiency. This paper presents a simple, versatile, and highly efficient procedure for the production of oligonucleotide-directed

mutations in DNA fragments cloned into M13 derived vectors. Identification of mutant DNA molecules is direct and does not depend on biological selection or multiple rounds of mutagenesis.

MATERIALS AND METHODS

E. coli JM101 and M13mp5 were the gift of J. Messing (University of Minnesota). E. coli DNA polymerase (large fragment) and T4 DNA ligase were obtained from Bethesda Research Laboratories. T4 polynucleotide kinase was purchased from New England Biolabs. α - 32 P-dATP and γ - 32 P-ATP were obtained from New England Nuclear. Deoxyribonucleotide triphosphates were purchased from P.L. Biochemicals. ATP, Trizma-base, and ultrapure sucrose were obtained from Sigma. The octadecanucleotide 5'-AAGGATAGCCITTAATC-3' was synthesized by the phosphormester method (M. Zoller, T. Atkinson, A. Markham and M. Smith, in preparation.) Hydrazine (99%) and acrylamide were purchased from Kodak. Dimethylsulfate was obtained from BDH. Formic acid (91%), liquified phenol, CaCl_2 , EDTA, and PEG-6000 were purchased from Fisher. Nitrocellulose filters were obtained from Schleicher and Schuell.

All of the mutagenesis reactions are carried out in 0.5 ml siliconized Eppendorf tubes and are mixed by gentle vortex or hand agitation. Additions of 5 μ l or less are made directly into the tube using siliconized 5 μ l graduated micropipets. The contents of the tube are kept at the bottom by a short spin (1-2 sec) in an Eppendorf centrifuge.

Computer analysis

DNA sequences were analyzed using the SEQUENCE program developed by A. Delaney (22).

Preparation of single stranded template DNA

A 4.2 kb Hind III fragment containing the yeast MATa gene was isolated as described in (23). The entire 4.2 kb fragment was ligated into the Hind III site of the vector M13mp5 (24). Approximately 13 ng of Hind III digested M13mp5 was mixed with 0.25 μ g of Hind III cleaved pYeMATa (pBR322 with a 4.2 kb Hind III fragment containing the MATa gene from yeast) in 50 mM tris-HCl pH 7.5, 10 mM MgCl_2 , 10 mM dithiothreitol, 1 mM ATP, and 0.4 units T4 DNA ligase in a total volume of 20 μ l. Ligation proceeded for 20 hr at 12° C. Aliquots of the ligation mixture were added to CaCl_2 treated E. coli JM101 cells as described in (25). Single stranded recombinant DNA was prepared from individual plaques by growing 1 ml phase cultures for 5 hr at 37° C (26). This short incubation

minimizes the production of deletions. A number of these recombinants were sequenced by the chain terminator method (27) near the insertion junction to determine the orientation of the MAT insert. One recombinant with the desired orientation was chosen and used as a source of single stranded DNA for the mutagenesis experiment. Template DNA from this recombinant was isolated from a 1 liter culture yielding approximately 3 mg of single stranded DNA. This DNA was stored in 1 ml of 10 mM tris-HCl 1 mM EDTA (pH 8) and served as a source for all mutagenesis experiments.

5' Phosphorylation of the oligonucleotide

A) For mutagenesis: Octadecanucleotide (200 pmol) was phosphorylated in a solution containing 0.1 M tris-HCl (pH 8), 10 mM $MgCl_2$, 5 mM DTT, 0.1 mM ATP and 4.5 units of T4 polynucleotide kinase in a total volume of 30 μ l (29). The reaction proceeded at 37° C for 45 min and was terminated by heating the sample at 65° C for 10 min.

B) For hybridization: Oligonucleotide (20 pmol) was phosphorylated as in A with 20 μ Ci γ - ^{32}P -ATP (2000 Ci/mmol) as the only source of ATP. The reaction was terminated as above then chromatographed on Sephadex G-25 (6 X 200 mm) in 50 mM ammonium bicarbonate (pH 7.8) to separate unincorporated ^{32}P -ATP from the ^{32}P -labeled oligonucleotide.

Aliquots (100 μ l) were collected in 1.5 ml Eppendorf tubes and ^{32}P in each fraction was determined by scintillation counting. Fractions containing the phosphorylated oligonucleotide (first radioactive peak) were pooled. The effectiveness of this separation was determined by chromatographing an aliquot (10,000 ^{32}P -cpm) on Whatman De-81 anion-exchange paper in 0.3 M ammonium formate (pH 8). Under these conditions the oligonucleotide remains at the origin and ATP migrates with an R_f of approximately 0.7. This procedure typically incorporated 4-8 X 10^6 ^{32}P -cpm into 20 pmol oligonucleotide.

Preliminary in vitro tests to determine specificity of priming

A) Primer extension: M13-recombinant DNA (0.5 pmol) was added to a 0.5 ml siliconized Eppendorf tube and mixed with 10 pmol of 5' ^{32}P -labeled oligonucleotide (2 X 10^4 cpm/pmol) in a solution containing 1 μ l Solution A (0.2 M tris-HCl pH 7.5 0.1 M $MgCl_2$, 0.5 M NaCl, and 0.01 M dithiothreitol) in a total volume of 10 μ l. The mixture was incubated at 55° C for 5 min then placed at 23° C. After 5 min, 1 μ l of dNTP Solution (2.5 mM of all four deoxyribonucleoside triphosphates) and 0.5 μ l of Solution A were added to the annealed mixture. Enzymatic

extension of the primer was initiated by the addition of 1 unit of DNA polymerase (large fragment). The reaction proceeded for 5 min at 23° C, 5 units of Hinf I were added and digestion proceeded for 1 hr at 37° C. An equal volume of formamide-dye solution (0.025% bromphenol blue, 0.025% Xylene cyanol FF, 0.01 M EDTA in 90% deionized formamide) was added and the mixture was heated in a boiling water bath for 3 min then chilled on ice. The fragments were electrophoresed on a 5% polyacrylamide-7 M urea gel (20 X 40 X 0.15 cm) using 0.09 M tris-borate (pH 8.3) 2.5 mM EDTA (28). Electrophoresis was carried out at 600 volts for 3 hr. The gel was autoradiographed using Kodak NS-5T film for 1-4 hr at 23° C.

B) Chain terminator sequencing: M13-recombinant DNA (0.1-0.3 pmol) was mixed with the mutagenic oligonucleotide (10-30X molar excess over template) and 1 µl of Solution A in a total volume of 10 µl. The solution was heated at 55° C for 5 min then placed at 23° C to cool for 5 min. Chain terminator sequencing was carried out according to Sanger *et al.* (26,27). Electrophoresis was conducted using a 10% polyacrylamide-7 M urea gel and proceeded for 3 hr at 1200 volts. The gel was covered with Saran wrap and autoradiographed for 12 hr at -20° C using Kodak NS-5T film.

Oligonucleotide-directed synthesis of covalently closed double stranded DNA

A) Annealing: Single stranded M13-recombinant DNA (1 pmol) was mixed with 5' phosphorylated oligonucleotide (20 pmol) and 1 µl Solution A in a total volume of 10 µl. The mixture was heated at 55° C for 5 min then placed at 23° C for 5 min.

B) Extension and ligation: 10 µl of Solution B (20 mM tris-HCl pH 7.5, 10 mM MgCl₂, 10 mM dithiothreitol, 1 mM dCTP, 1 mM dTTP, 1 mM dGTP, 1mM riboATP, 5 µM α-³²P-dATP (200 Ci/mmol), 3 units T4 DNA ligase were added to the tube containing the annealed DNA. 2.5 units of *E. coli* DNA polymerase (large fragment) were added and the solution was incubated for 5 min at 23° C. 1 µl 10 mM dATP was added and the sample was incubated at 15° C for 20 hr (unless otherwise indicated).

Enrichment for covalently closed double stranded DNA

Following extension and ligation unincorporated α³²P-dATP was removed from the *in vitro* synthesized DNA by precipitation of the DNA with a PEG/NaCl solution. 30 µl water and 50 µl 13% PEG-6000/1.6 M NaCl were added to the reaction tube, which was placed on ice for 15 min. The

DNA was pelleted by centrifugation for 5 min in an Eppendorf centrifuge and the supernatant was removed using a disposable micropipet. 100 μ l of cold 6.5% PEG/0.8 M NaCl solution was added to rinse down the sides of the tube. The solution was collected by a 30 sec Eppendorf spin and removed. The precipitated DNA was resuspended in 180 μ l of 10 mM tris-HCl (pH 8) 1 mM EDTA. 20 μ l of 2 N NaOH were added, the sample was incubated at 23° C for 5 min, and then chilled on ice for 1 min. The sample was applied to a 5–20% alkaline sucrose gradient that contained 0.2 N NaOH, 1 M NaCl, and 2 mM EDTA. Centrifugation was carried out at 37,000 rpm for 2 h at 4° C using an SW50.1 rotor. Following centrifugation, aliquots (175 μ l) were collected by puncturing the centrifuge tube at the bottom and 32 P-cpm of each fraction was determined by scintillation counting. The fractions containing closed circular double stranded DNA (faster migrating peak) were pooled and neutralized to pH 8 with 1 M tris-citrate (pH 5). Approximately 50 μ l of 1 M tris-citrate were added to 300 μ l gradient solution.

Transformation using CC-DNA

Aliquots of the pooled DNA (1, 2, 5, and 10 μ l) were used to transform *CaCl*₂ treated *E. coli* JM101 cells (0.2 ml) (30). Approximately 100–200 plaques were obtained per 1 μ l CC-DNA solution.

Preparation of recombinant phage DNA for screening

Recombinant phage from 36 individual plaques were prepared by removing the entire plaque from the plate with a disposable micropipet and adding it to 1 ml of early log phase JM101 in 2X YT (1.6% tryptone, 1.0% yeast extract, 0.5% NaCl, and 0.1% dextrose). The cultures were incubated with vigorous shaking for 5 hr at 37° C. Single stranded DNA was isolated (26) from the phage containing supernatant and precipitated at –70° C with 0.1 volume 3 M sodium acetate and 3 volume ethanol. Following ethanol precipitation, the DNA was dried *in vacuo* and resuspended in 50 l 10 mM tris-HCl (pH 8) 0.1 mM EDTA.

Screening procedures

A) Primer extension and restriction digestion to detect loss of Hinf I site as a result of the mutation: 5 μ l M13-recombinant DNA (0.2 pmol) were mixed with 1 μ l of Solution A and 0.2 pmol of 0.57 kb Xba I restriction fragment from MATa in a total volume of 10 μ l. The mixture was heated at 100° C for 3 min then placed at 67° C for 30 min to hybridize. (Alternatively, 1.5 pmol of an M13 sequencing primer can be substituted for the restriction fragment primer. In this case, the

reaction vial is heated at 55° C for 5 min then placed at 23° C for 5 min to cool.) Once the sample had cooled, 1 µl each of 0.5 mM dCTP, 0.5 mM dTTP, and 0.5 mM dGTP, 1 µl 0.05 mM dATP 5 µCi α-³²P-dATP (2000 Ci/mmol), and 0.5 µl Solution A were added to the mixture. Primer extension was initiated by addition of 1 unit of DNA polymerase (large fragment) and proceeded at 23° C for 10 min. 1 µl 2.5 mM dNTP Solution (all four deoxyribonucleoside triphosphates each at a concentration of 25 mM) was added and reaction continued for another 5 min at 23° C. The reaction was terminated by heating the sample at 65° C for 10 min. After cooling to 37° C, 5 units of Hinf I were added and digestion proceeded at 37° C for 1 hr. 5 µl sucrose-dye mix (60% sucrose, 0.025% bromphenol blue, 0.025% Xylene cyanol FF, and 25 mM EDTA) were added and the entire sample was applied to a 5% non-denaturing polyacrylamide gel (20 X 40 X 0.15 cm). Electrophoresis was carried out at 200 volts for 12 hr using 50 mM tris-borate (pH 8.3) 1 mM EDTA (28). The gel was autoradiographed for 3 hr at 23° C using Kodak NS-5T film.

B) Dot blot hybridization using the mutagenic oligonucleotide as probe: 1 µl of single stranded recombinant DNA (out of 50 µl total) from each isolate was spotted on a dry sheet of nitrocellulose (10 X 10 cm). The filter was baked at 80° C in vacuo for 2 hr, then prehybridized in 5 ml 6X SSC (1X SSC = 0.15 M NaCl, 0.015 M sodium citrate, 1 mM EDTA, (pH 7.2) + 10X Denhardt's solution (100X = 2% bovine serum albumin, 2% polyvinyl pyrrolidone, 2% Ficoll) for 15 min at 23° C in a sealed cooking bag. The prehybridization solution was removed, 4 ml probe solution was added, and the cooking bag was resealed. The probe solution consisted of 1 X 10⁶ ³²P-cpm (0.3 pmol) of 5' phosphorylated oligonucleotide added to 4 ml 6X SSC + 10X Denhardt's solution. Hybridization was carried out at 23° C for 1 hr. The filter was washed with 3 X 50 ml of 6X SSC at 23° C using Kodak NS-5T film. The filter was washed again with 50 ml 6X SSC for 5 min at 37° C followed by autoradiography for 1 hr. This was repeated with a 47° C wash for 5 min. The filter was autoradiographed for 1 hr then for 8 hr.

Isolation of double stranded mutant DNA

Mutant phage were plated onto YT plates (0.8% tryptone, 0.5 yeast extract, 0.5% NaCl, 0.1% dextrose, and 1.5% agar) to yield isolated plaques. A single plaque was removed with a disposable micropipet, added to 1 ml YT media, and incubated with shaking at 37° C. After 4 hr, the entire 1 ml culture and 10 ml of fresh exponentially growing JM101 cells

were added to 1 liter of YT. This was incubated at 37° C until it reached a density with an A600= 0.75. The cells were harvested by centrifugation at 5000 rpm for 5 min and double-stranded phage DNA was isolated from a cleared lysate (31). The DNA was subjected to two CsCl isopycnic centrifugations in the presence of ethidium bromide, dialyzed for 16 hr versus 3 X 1 liter of 10 mM tris-HCl (pH 8) 1 mM EDTA at 4° C, ethanol precipitated, and resuspended in 0.5 ml 10 mM tris-HCl (pH 8) 1 mM EDTA. The yield from 1 litre was approximately 0.25 mg.

DNA sequence determination

Restriction fragments prepared from double stranded mutant DNA were 3' end-labeled, electrophoresed in a 5% non-denaturing polyacrylamide gel, and electroeluted from the gel (28). DNA sequence determination was carried out according to the procedure of Maxam and Gilbert (2). The sequence of the oligonucleotide was determined using a modification of this procedure. 5' phosphorylated oligonucleotide (500,000 ³²P-cpm, 20 pmol) was subjected to the following modification reactions: the 'C' and 'C+T' reactions were carried out at 55° C instead of 23° C using 30 µl hydrazine for 15 min, the 'A+G' reaction consisted of the addition of 3 µl 50% formic acid followed by incubation at 37° C for 25 min, and the 'G' reaction was carried out at 37° C using 2 µl dimethylsulfate for 15 min. All other modification and subsequent cleavage reactions were followed as outlined in (28). The ethanol precipitation steps following the modification reactions inefficiently precipitates short oligonucleotides and results in a significant loss of ³²P-cpm. This has been allowed for by using more ³²P-labeled oligonucleotide than usually required.

RESULTS

The DNA sequence of MATa1 (32) in the region containing an inframe TGA codon is shown in Figure 2. The octadecanucleotide was designed to change the TGA to TAA. Prior to the synthesis of the oligonucleotide a computer analysis (22) was carried out for sequences in the + strand of the vector M13mp5 (33) and the MATa insert that were complementary to the oligonucleotide. The purpose of this analysis was to identify competing sites from which oligonucleotide-directed DNA synthesis might also occur.

Before conducting the actual mutagenesis experiment, two preliminary tests were performed that demonstrated specific priming by the mutagenic oligonucleotide at the target site. First, 5' ³²P-labeled

^{H F K D S L ■ I N}
 5'-TTTCATTTCAGGATAGCCTTTGAATCAATTA-3' coding strand
 Hinf I
 ●
 5'-AAGGATAGCCTTTAAATC-3' mutagenic oligonucleotide

Figure 2. The DNA sequence of the coding strand of MATa1 in the region containing the inframe TGA codon. Below is shown the sequence of the oligonucleotide synthesized to change the TGA to TAA. The mutation destroys the Hinf I site in this region (underlined).

oligonucleotide was annealed with the template DNA then extended by the action of E. coli DNA polymerase (large fragment). Following extension, the in vitro synthesized double stranded DNA was cleaved with Hinf I. The resulting fragments were denatured and electrophoresed on a 5% polyacrylamide-7 M urea gel then autoradiographed. Since the DNA sequence of the MATa insert was known, and hence the distance from the desired priming site to the first downstream Hinf I site, the exact size of the fragment that would be produced by correct priming at the desired site could be predicted. Correct priming would produce a fragment consisting of 300 nucleotides. A 300 nucleotide fragment was the predominant product based on the intensity of the radiographic signal.

In order to directly show that the major priming site observed by primer extension was the desired target site, the mutagenic oligonucleotide was used as a sequencing primer in a chain terminator reaction. The sequence obtained corresponded to the sequence in MATa1 approximately 30 bases downstream from the TGA. The clarity of the pattern suggested that priming specifically occurred at the desired site.

The mutagenesis procedure is outlined in Figure 1. A potential problem in oligonucleotide-directed mutagenesis is the background of wild type molecules resulting from the inefficient conversion of single stranded template DNA into double stranded CC-DNA (10,11). In experiments with ØX174, digestion with S1 nuclease following the extension and ligation reactions substantially increased the percentage of mutants obtained (11). Other useful methods include binding to nitrocellulose (10), agarose gel electrophoresis (16), CsCl density gradient centrifugation (34), and alkaline sucrose gradient centrifugation (17). We chose alkaline sucrose gradient centrifugation because it efficiently enriches for doubled stranded CC-DNA, it provides a diagnostic indication of the efficiency of the extension and ligation

reactions, and it is a simple procedure.

Three methods can be used to screen for mutant DNA, all of which use a small aliquot of the DNA isolated from a 1 mM culture. 1) Chain terminator sequencing using the dideoxynucleotide that distinguishes the alteration, 2) primer extension and restriction endonuclease digestion, or 3) DNA-oligonucleotide hybridization on nitrocellulose using the mutagenic oligonucleotide as probe. The first method is convenient if the mutation can be sequenced using a universal M13 sequencing primer or if another oligonucleotide or restriction fragment that primes adjacent to the target site is available. The second method can be used if the mutation creates or destroys a restriction site. The third method is independent of the position of the mutation, can be conducted using either intact phage or isolated single stranded DNA, and identifies the mutant in a single day. The principle behind the hybridization method is that the mutagenic oligonucleotide will form a more stable duplex with mutant DNA, with perfect match, than with wild type DNA, bearing a mismatch (6,7,35,36). Hybridization is initially carried out at conditions of low stringency where both wild type and mutant DNA hybridize with the ^{32}P -labeled oligonucleotide. Mutants are detected by washing the filter at increasingly higher temperatures until only mutant molecules hybridize. This stepwise scanning through the melting temperature of oligonucleotide duplexes is extremely rapid and requires no empirical assumptions about melting temperature.

The same 36 M13-recombinants were screened by the hybridization method using ^{32}P -labeled oligonucleotide as the probe (Figure 3). The filter was washed in 6X SSC at 23° C, 37° C, and 47° C. Autoradiography was carried out after each wash for 1 hr (panels A-C) and 8 hr (panel D). Four clones exhibited strong signals after the 47° C wash. These corresponded to the same recombinants that exhibited the new 650 bp Hinf I fragment.

Phage from two of the suspected mutants (7 and 32) were plated out and single stranded DNA from 16 individual plaques was isolated and rescreened for the mutation by the restriction digest method. All recombinants exhibited the new 650 bp fragment. However, in other experiments conducted in our laboratory, we have observed that suspected mutants can contain up to 50% wild type molecules. Therefore, it is important to plaque purify the mutant.

Double stranded DNA was prepared from one mutant phage and sequenced

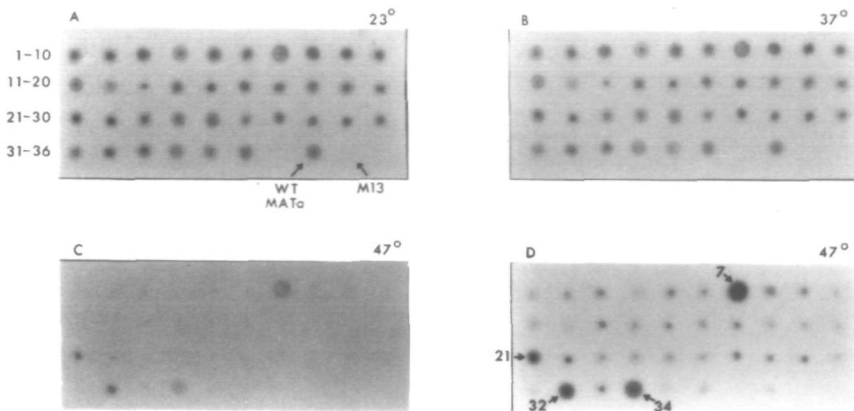


Figure 3. Dot blot hybridization screening for mutants using ^{32}P -labeled mutagenic oligonucleotide. Single stranded DNA from 36 phage was bound to nitrocellulose and hybridized to ^{32}P -labeled oligomer. Hybridization was carried out 23°C for 1 hr. The filter was washed in 6X SSC at 23° , 37° , and 47°C , for 5 min and autoradiographed for 1 hr (panels A-C) or 8 hr (panel D) using Kodak NS-5T film.

by the method of Maxam and Gilbert. The only difference between the two sequences was a single C to T transition at the desired site (Figure 4).

DISCUSSION

This paper presents a procedure for oligonucleotide-directed mutagenesis that is versatile, simple and highly efficient. The development of the method was guided by three criteria: 1) that any point mutation could be produced at any position in a cloned fragment of

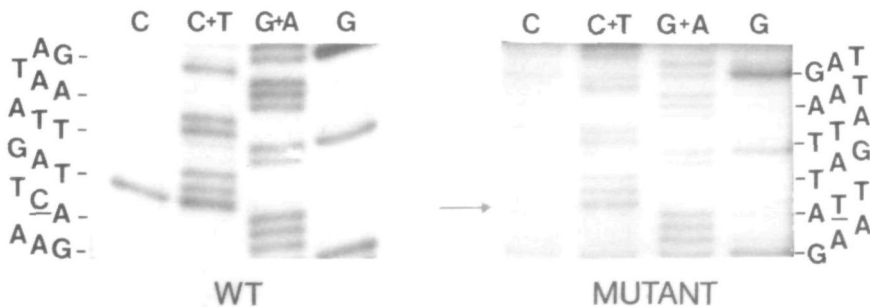


Figure 4. Comparison of mutant and wild type DNA sequences by the Maxam and Gilbert procedure. The difference between the two is the desired C to T change (arrow).

known sequence, 2) that the efficiency of mutagenesis would be sufficiently high to facilitate screening, and 3) that the identification of mutant molecules could be made without biological selection or screening.

The construction of a defined point mutation in a clone of the MATa1 gene is described. The efficiency of mutagenesis was 11%. The generality of this procedure has been demonstrated by the successful completion of 8 additional mutagenesis experiments conducted in our laboratory with efficiencies of 6-45% (A. Spence, S. Porter, S. Goh, D. Russell, G. Winter, and M. Zoller, unpublished). The size of the cloned DNA fragments used in these experiments ranged from 250 bases to 4.2 kilobases and the mutagenic oligonucleotides were 14-16 nucleotides long.

A number of points merit emphasis. First, it is important to conduct a thorough computer search for competing sites before synthesizing the mutagenic oligonucleotide. Second, the preliminary tests to demonstrate specific priming at the target site should be carried out. These tests are quite simple and yield important information. Third, the alkaline sucrose gradient step is essential in that it not only serves to enrich for double stranded CC-DNA, but it also gives a clear indication of the efficiency and kinetics with which CC-DNA molecules are formed. Fourth, we have observed considerable variation among the efficacies of DNA polymerase (large fragment) from different suppliers. For example, in parallel experiments using different E. coli DNA polymerase I (large fragment) preparations only one out of three yielded a population of CC-DNA molecules. In all cases, however, the same amount of α -³²P-dATP was incorporated during the initial period of extension (5 min at 23° C). This variability was also observed by Baas et. al. (34). The source of the problem is unknown. However, these examples point out the usefulness of the alkaline sucrose gradient as an assay for the production of double stranded closed circular molecules.

It is significant to note that, in general, the efficiencies of mutagenesis of fragments cloned into M13 vectors using the procedure described in this paper are similar to the efficiencies of mutagenesis of ϕ 174 (1). This suggests that mismatch repair systems (37) are not a serious problem as might have been concluded from other oligonucleotide-directed mutagenesis studies in M13 vectors or pBR322 (13-21). Rather, the major problems appear to lie in the inefficient production of double

stranded CC-DNA and in the variable quality of *E. coli* DNA polymerase (large fragment) preparations. The procedures described in this paper allow one to eliminate these problems.

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